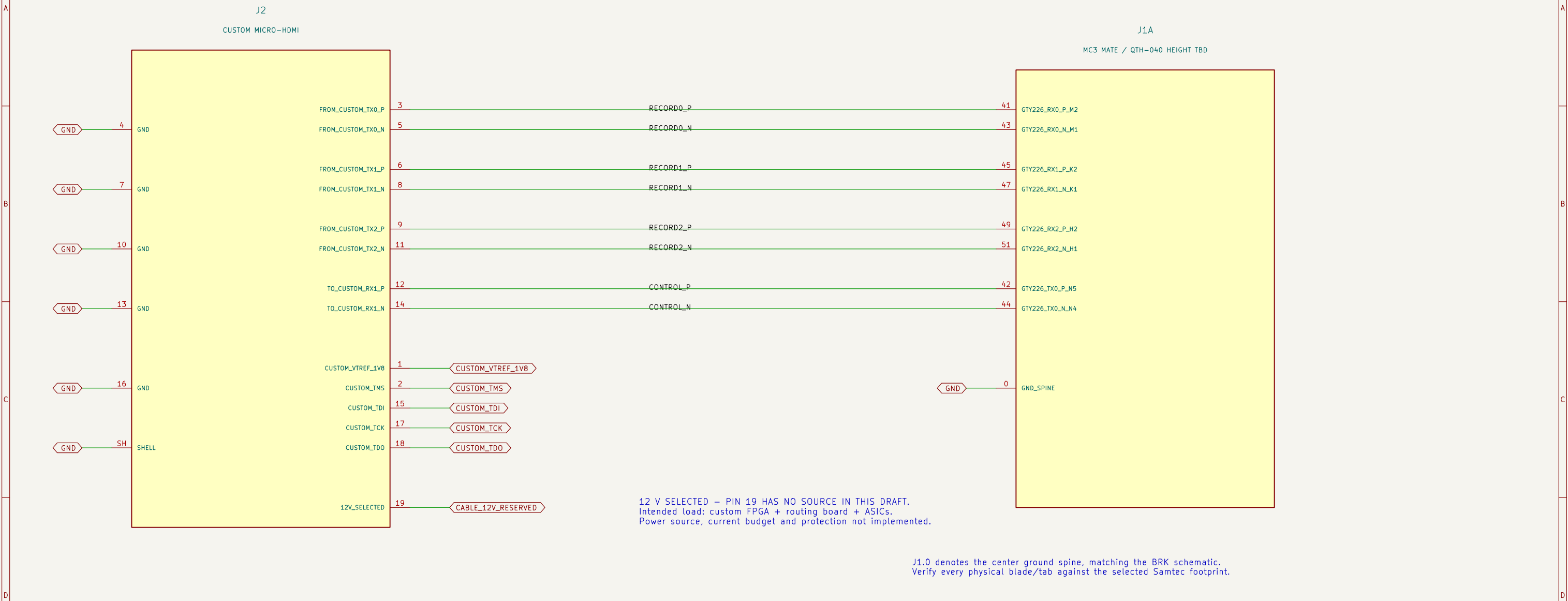


XEM8310 adapter R2 / 01 – recording and command cable

Direct MC3 carrier. Three recording pairs to XEM; one control pair to custom FPGA. Custom cable, not HDMI video.



AC coupling: custom FPGA board already has C137–C144, 100 nF per conductor.
No second set is added here. No external termination is added; GT configuration must match.
Clock: XEM onboard 125 MHz reference at bank 225 is proposed for bank 226 via internal routing; Vivado proof pending.

Draft shell treatment: connector shell bonded to common GND. Cable shield/ESD details require review.
Mating footprints, stack height, differential layout and signal integrity are not defined by this schematic.

02 / separate custom-FPGA and XEM JTAG

03 / unused MC1, MC2 and MC3 contacts

Source: pins.opalkelly.com/pin_list/XEM8310.pdf
Official pin numbers identify XEM-facing contacts, not a PCB view.

CONNECTION DRAFT / NOT FOR FABRICATION

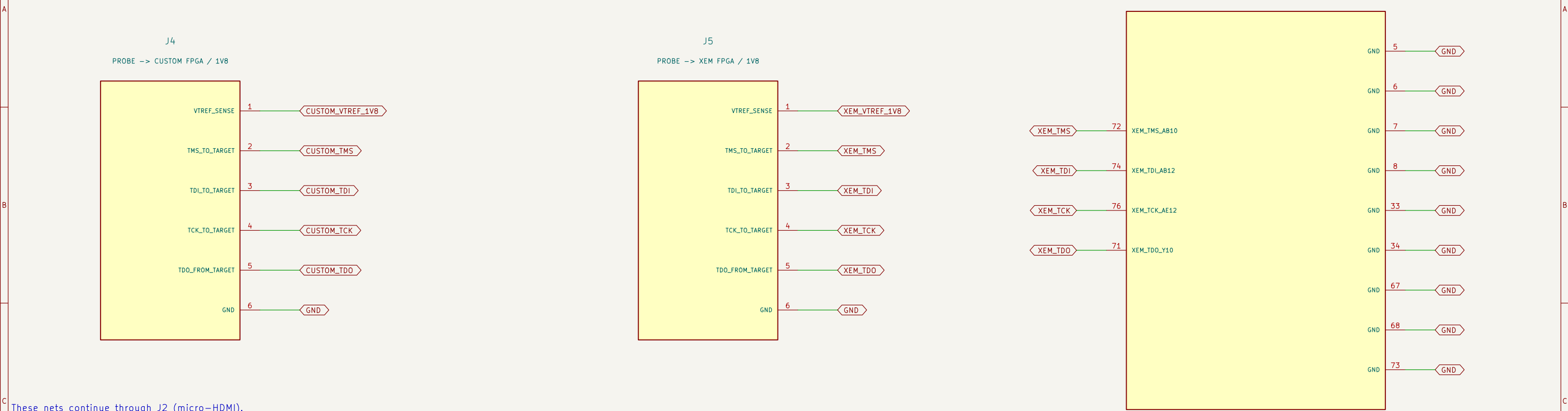
Sheet: /
File: XEM8310_MicroHDMIAdapter_R2_12V.kicad_sch

Title: XEM8310 adapter R2 / 01 – recording and command cable

Size: A3 Date: 2026-09-29 Rev: R2 REVIEW
KiCad E.D.A. 10.0.6 Id: 1/3

XEM8310 adapter R2 / 02 – two independent JTAG targets

Generic 6–contact probe connections: pin order is explicitly shown. No direct plug compatibility is claimed.



These nets continue through J2 (micro–HDMI).
VTREF is supplied by the powered custom FPGA board.
Probe senses VTREF; it must not source this rail.
Custom JTAG TDO already has R118 (33 ohm) at target.

XEM JTAG is 1.8 V. General 1.8 V reference comes from MC1.2.
TMS / TDI / TCK have 4.7 kohm pull–ups on XEM.

J4 and J5 are separate JTAG targets; only GND is shared.
Generic pin order: NOT directly plug–compatible with HS2/HS3.
VTREF is target–supplied; intended for probe reference only.
Selected probe loading on VTREF must be verified.
FrontPanel USB configures XEM; JTAG enables Vivado debug.

XEM uses its own intended external power input.
12 V cable supply selected; source and protection not implemented.

Sources: docs.opalkelly.com/xem8310/jtag–and–sysmon/ | expansion–connectors/
BRK8310–A01–Schematic.PDF confirms MC1.2 = general 1.8 V; MC2.75 VCCADC is not used.

CONNECTION DRAFT / NOT FOR FABRICATION

Sheet: /02 / separate custom–FPGA and XEM JTAG/
File: JTAG_Connections.kicad_sch

Title: XEM8310 adapter R2 / 02 – two independent JTAG targets

Size: A3	Date: 2026–09–29	Rev: R2 REVIEW
KiCad E.D.A. 10.0.6		Id: 2/3

XEM8310 adapter R2 / 03 – unused carrier contacts

These are the remaining units of the SAME physical J1, J3 and J6 connectors. NC means no carrier trace in this draft.

J1B				J3B				J6B			
MC3 MATE / QTH-040 HEIGHT TBD				MC2 MATE / BTE-040 HEIGHT TBD				MC1 MATE / BTE-040 HEIGHT TBD			
✕1	UNUSED_01	UNUSED_37	✕37	✕1	UNUSED_01	UNUSED_41	✕41	✕1	UNUSED_01	UNUSED_41	✕41
✕2	UNUSED_02	UNUSED_38	✕38	✕2	UNUSED_02	UNUSED_42	✕42	✕4	UNUSED_04	UNUSED_42	✕42
✕3	UNUSED_03	UNUSED_39	✕39	✕3	UNUSED_03	UNUSED_43	✕43	✕5	UNUSED_05	UNUSED_43	✕43
✕4	UNUSED_04	UNUSED_40	✕40	✕4	UNUSED_04	UNUSED_44	✕44	✕6	UNUSED_06	UNUSED_44	✕44
✕5	UNUSED_05	UNUSED_46	✕46	✕9	UNUSED_09	UNUSED_45	✕45	✕7	UNUSED_07	UNUSED_45	✕45
✕6	UNUSED_06	UNUSED_48	✕48	✕10	UNUSED_10	UNUSED_46	✕46	✕8	UNUSED_08	UNUSED_46	✕46
✕7	UNUSED_07	UNUSED_50	✕50	✕11	UNUSED_11	UNUSED_47	✕47	✕9	UNUSED_09	UNUSED_47	✕47
✕8	UNUSED_08	UNUSED_52	✕52	✕12	UNUSED_12	UNUSED_48	✕48	✕10	UNUSED_10	UNUSED_48	✕48
✕9	UNUSED_09	UNUSED_53	✕53	✕13	UNUSED_13	UNUSED_49	✕49	✕11	UNUSED_11	UNUSED_49	✕49
✕10	UNUSED_10	UNUSED_54	✕54	✕14	UNUSED_14	UNUSED_50	✕50	✕12	UNUSED_12	UNUSED_50	✕50
✕11	UNUSED_11	UNUSED_55	✕55	✕15	UNUSED_15	UNUSED_51	✕51	✕13	UNUSED_13	UNUSED_51	✕51
✕12	UNUSED_12	UNUSED_56	✕56	✕16	UNUSED_16	UNUSED_52	✕52	✕14	UNUSED_14	UNUSED_52	✕52
✕13	UNUSED_13	UNUSED_57	✕57	✕17	UNUSED_17	UNUSED_53	✕53	✕15	UNUSED_15	UNUSED_53	✕53
✕14	UNUSED_14	UNUSED_58	✕58	✕18	UNUSED_18	UNUSED_54	✕54	✕16	UNUSED_16	UNUSED_54	✕54
✕15	UNUSED_15	UNUSED_59	✕59	✕19	UNUSED_19	UNUSED_55	✕55	✕17	UNUSED_17	UNUSED_55	✕55
✕16	UNUSED_16	UNUSED_60	✕60	✕20	UNUSED_20	UNUSED_56	✕56	✕18	UNUSED_18	UNUSED_56	✕56
✕17	UNUSED_17	UNUSED_61	✕61	✕21	UNUSED_21	UNUSED_57	✕57	✕19	UNUSED_19	UNUSED_57	✕57
✕18	UNUSED_18	UNUSED_62	✕62	✕22	UNUSED_22	UNUSED_58	✕58	✕20	UNUSED_20	UNUSED_58	✕58
✕19	UNUSED_19	UNUSED_63	✕63	✕23	UNUSED_23	UNUSED_59	✕59	✕21	UNUSED_21	UNUSED_61	✕61
✕20	UNUSED_20	UNUSED_64	✕64	✕24	UNUSED_24	UNUSED_60	✕60	✕22	UNUSED_22	UNUSED_62	✕62
✕21	UNUSED_21	UNUSED_65	✕65	✕25	UNUSED_25	UNUSED_61	✕61	✕23	UNUSED_23	UNUSED_63	✕63
✕22	UNUSED_22	UNUSED_66	✕66	✕26	UNUSED_26	UNUSED_62	✕62	✕24	UNUSED_24	UNUSED_64	✕64
✕23	UNUSED_23	UNUSED_67	✕67	✕27	UNUSED_27	UNUSED_63	✕63	✕25	UNUSED_25	UNUSED_65	✕65
✕24	UNUSED_24	UNUSED_68	✕68	✕28	UNUSED_28	UNUSED_64	✕64	✕26	UNUSED_26	UNUSED_66	✕66
✕25	UNUSED_25	UNUSED_69	✕69	✕29	UNUSED_29	UNUSED_65	✕65	✕27	UNUSED_27	UNUSED_67	✕67
✕26	UNUSED_26	UNUSED_70	✕70	✕30	UNUSED_30	UNUSED_66	✕66	✕28	UNUSED_28	UNUSED_68	✕68
✕27	UNUSED_27	UNUSED_71	✕71	✕31	UNUSED_31	UNUSED_69	✕69	✕31	UNUSED_31	UNUSED_69	✕69
✕28	UNUSED_28	UNUSED_72	✕72	✕32	UNUSED_32	UNUSED_70	✕70	✕32	UNUSED_32	UNUSED_70	✕70
✕29	UNUSED_29	UNUSED_73	✕73	✕35	UNUSED_35	UNUSED_75	✕75	✕33	UNUSED_33	UNUSED_71	✕71
✕30	UNUSED_30	UNUSED_74	✕74	✕36	UNUSED_36	UNUSED_77	✕77	✕34	UNUSED_34	UNUSED_72	✕72
✕31	UNUSED_31	UNUSED_75	✕75	✕37	UNUSED_37	UNUSED_78	✕78	✕35	UNUSED_35	UNUSED_73	✕73
✕32	UNUSED_32	UNUSED_76	✕76	✕38	UNUSED_38	UNUSED_79	✕79	✕36	UNUSED_36	UNUSED_74	✕74
✕33	UNUSED_33	UNUSED_77	✕77	✕39	UNUSED_39	UNUSED_80	✕80	✕37	UNUSED_37	UNUSED_75	✕75
✕34	UNUSED_34	UNUSED_78	✕78	✕40	UNUSED_40			✕38	UNUSED_38	UNUSED_76	✕76
✕35	UNUSED_35	UNUSED_79	✕79					✕39	UNUSED_39	UNUSED_79	✕79
✕36	UNUSED_36	UNUSED_80	✕80					✕40	UNUSED_40	UNUSED_80	✕80

Unused GTY RX: omit the channel instance or set RXPd = 2'b11. Do not enable a floating receiver.
Unused TX and external reference clock pairs are NC; do not instantiate IBUFDS_GTE4 for an unused reference input.
All three XEM underside connectors are now represented. MC3 bank 67 and other unused fabric I/O are NC.

Unused GTY guidance: AMD UG578 v1.4, PCB checklist (p340). Footprints and mechanical drawing remain pending.

CONNECTION DRAFT / NOT FOR FABRICATION		
Sheet: /03 / unused MC1, MC2 and MC3 contacts/ File: Unused_Connector_Contacts.kicad_sch		
Title: XEM8310 adapter R2 / 03 – unused carrier contacts		
Size: A3	Date: 2026-09-29	Rev: R2 REVIEW
KiCad E.D.A. 10.0.6		Id: 3/3